Towards an Explainable Conversational Medication Recommender System For Clinical Decision Support

AHMAD Adeel[1], AHMED Zaheer[2]

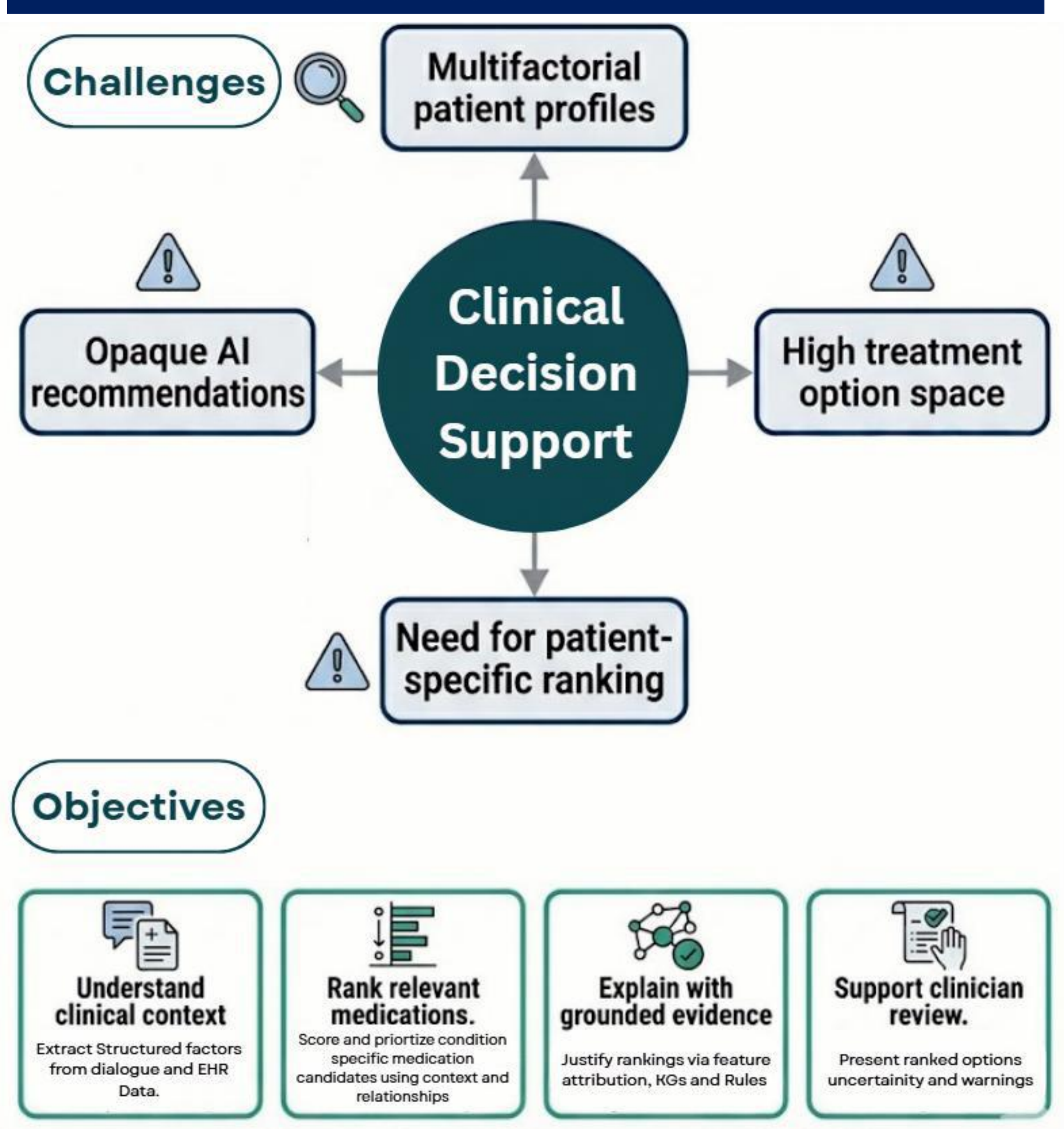
SysReIC, Laboratoire d'Informatique Signal et Image de la Côte d'Opale,
Université du Littoral Côte d'Opale, Calais, France

[1]adeel.ahmad@univ-littoral.fr, [2]dev.zaheer.ahmad@gmail.com

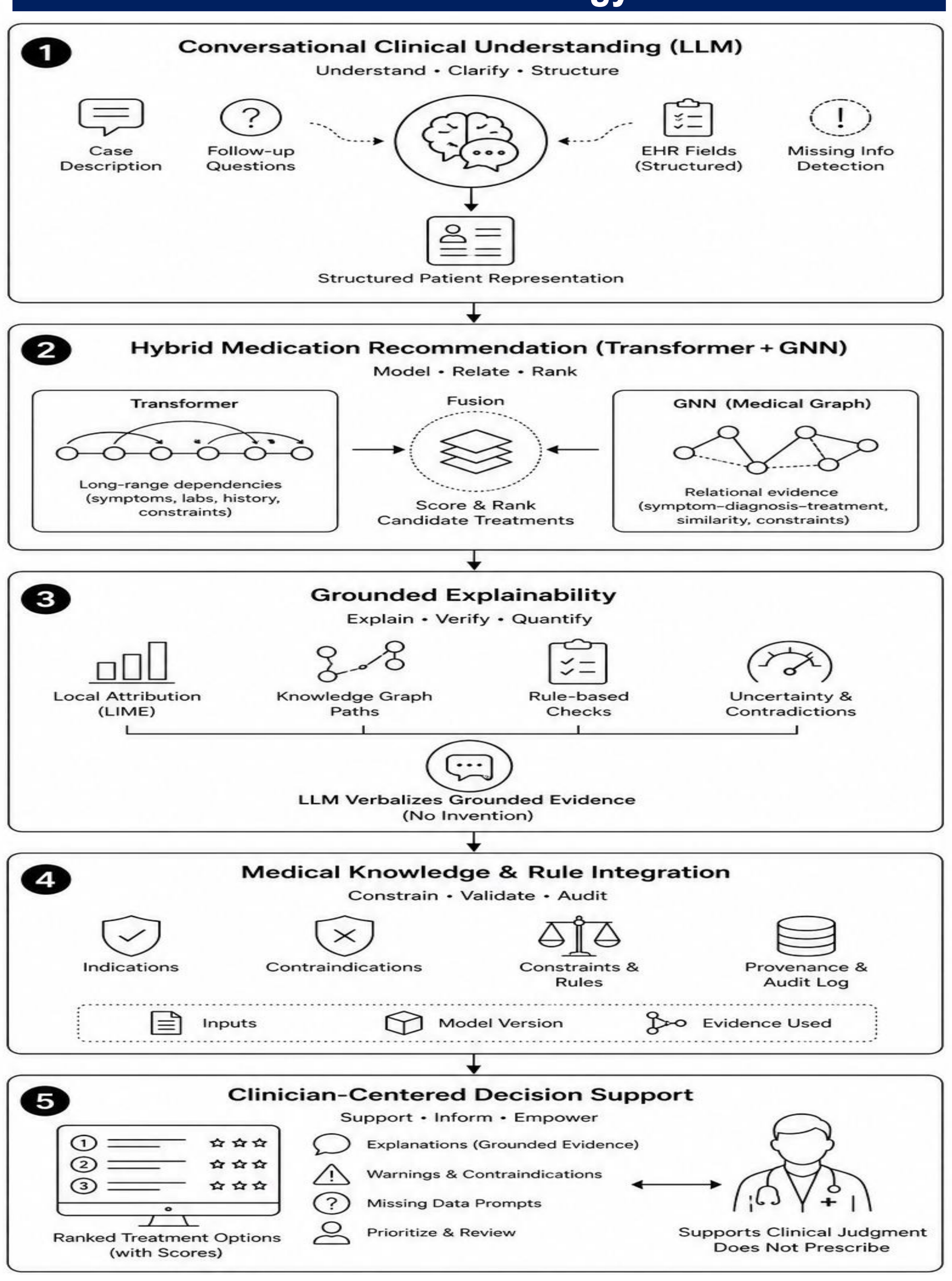
Objective

To develop an explainable conversational medication recommender for clinical decision support. The system will help healthcare professionals evaluate and prioritize medication options by (1) understanding clinical dialogue and structured records, (2) ranking medication candidates with a hybrid recommendation module that models both contextual dependencies and medical relationships, and (3) explaining each ranking with grounded evidence from patient features, graph paths, and clinical rules. By separating recommendation from explanation, the research addresses the lack of faithful, clinically understandable justification in many healthcare AI systems .

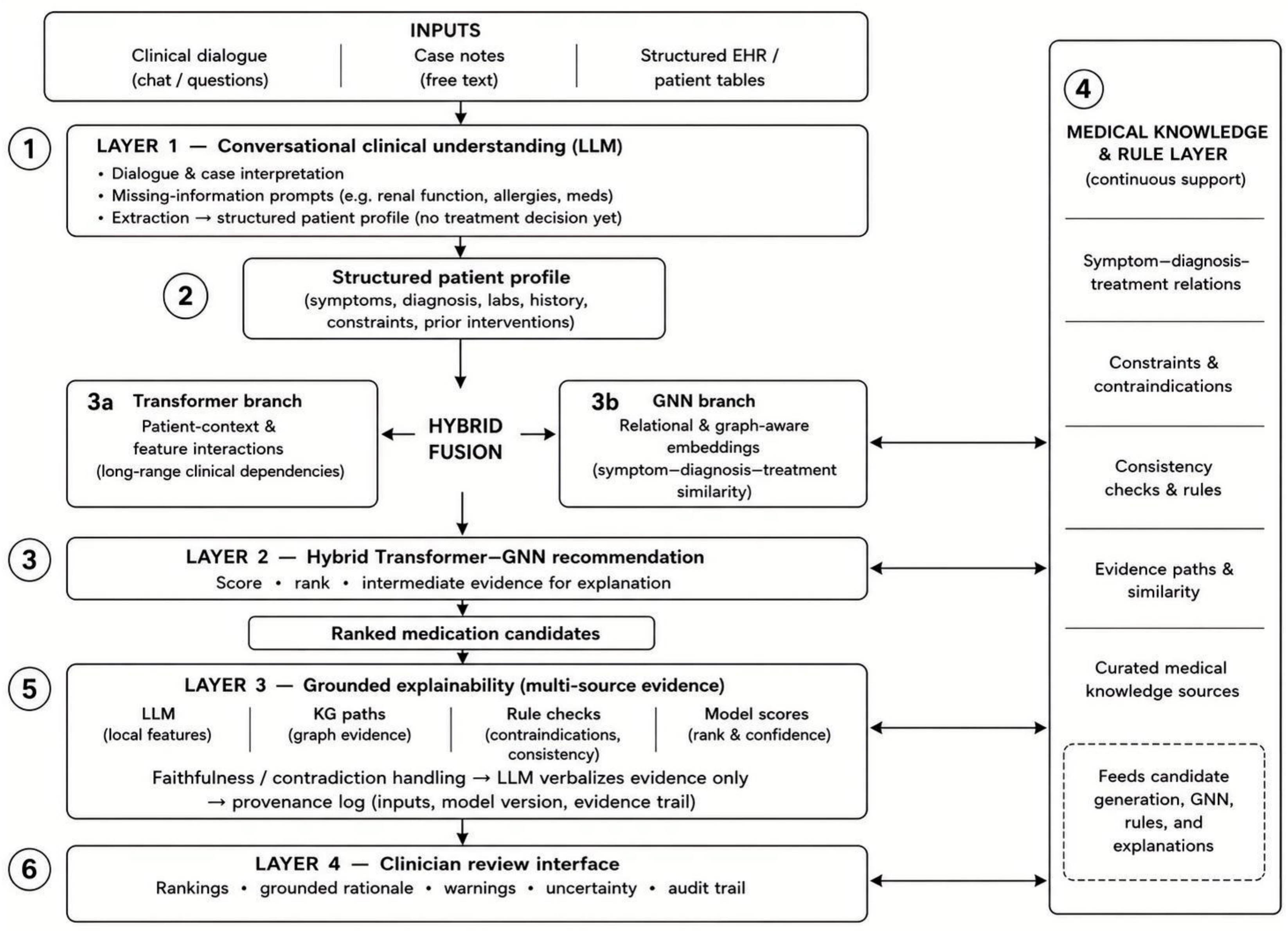
Problematic & Motivation



Solution Strategy



System Architecture



1. Conversational clinical understanding (LLM layer) :
Interprets clinician questions and case narratives; requests missing data; maps free text and EHR fields into a structured patient profile.

2. Structured patient representation & preprocessing:

- Integrates conversational extractions with structured records
- Normalizes demographics, diagnoses, comorbidities, and laboratory values into a single patient-context representation for ranking and audit.

3. Hybrid Transformer–GNN recommendation engine:

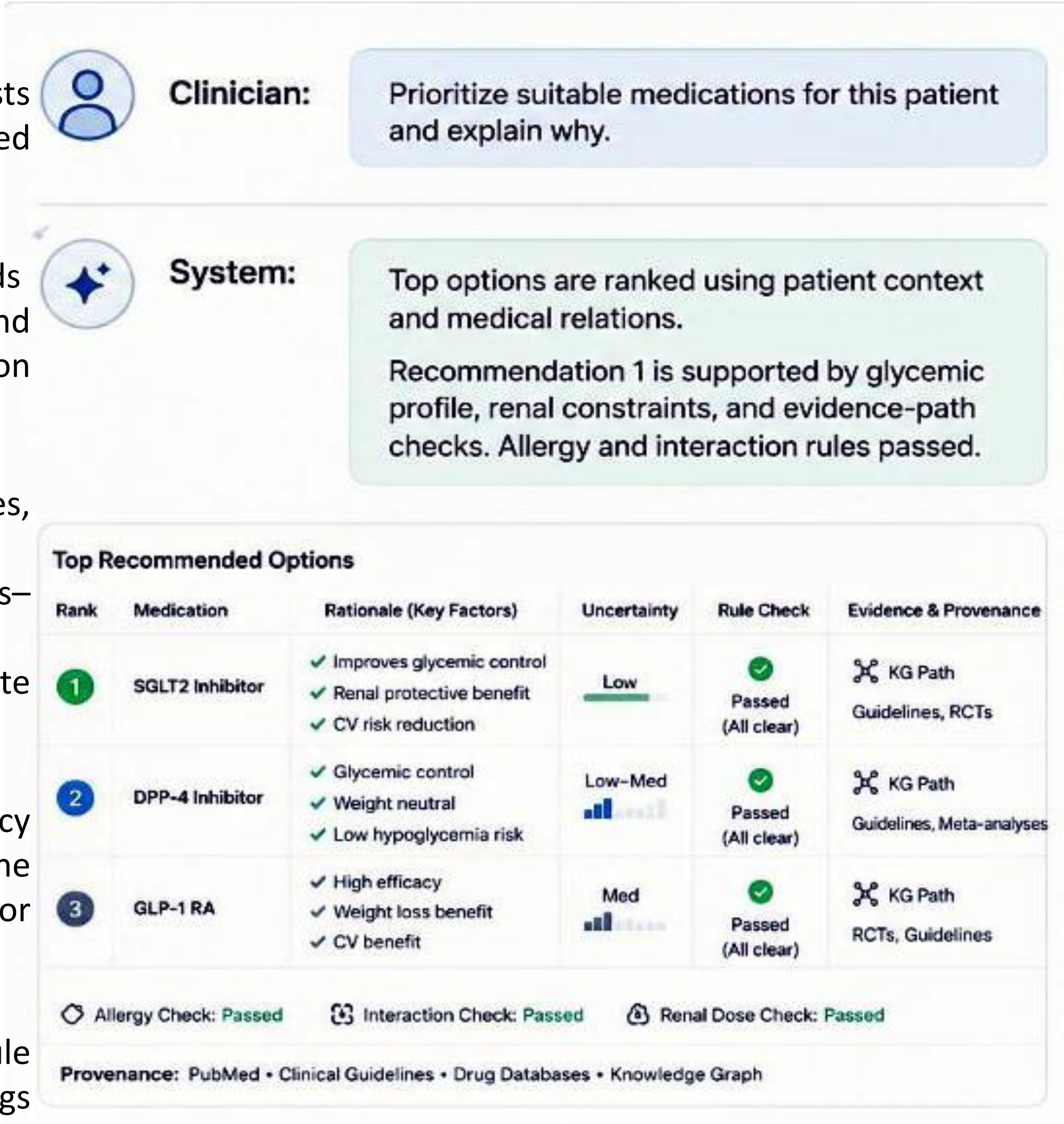
- **Transformer:** models interactions across patient features, history, and clinical context.
- **GNN:** models relational structure (symptom–diagnosis–treatment, patient similarity, constraints).
- **Fusion:** produces ranked medications plus intermediate evidence artifacts for explanation.

4. Medical knowledge & rule-based reasoning layer:
Supplies explicit relations, contraindication and consistency checks, and graph evidence paths used by both the recommender and the explainability module; can override or flag high scores when rules conflict.

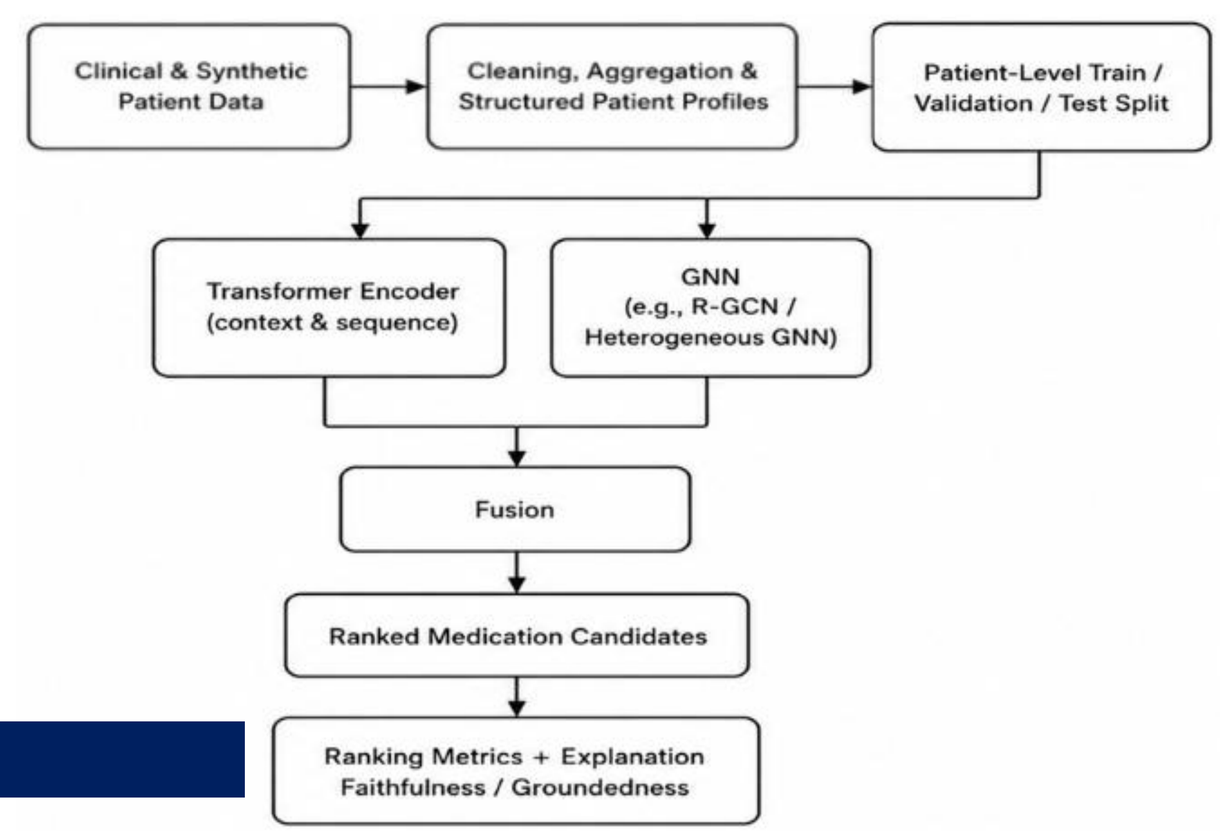
5. Grounded explainability module:
Combines LIME attribution, knowledge-graph paths, rule outcomes, and model scores; detects contradictions; logs provenance; uses the LLM as an **evidence verbalizer**, not as the source of truth.

Functional Scenario: Conversational, Evidence-Grounded Medication Prioritization

A clinician submits a complex patient case (diagnosis, symptoms, lab trends, comorbidities, current medications, and constraints) through the conversational interface. The system extracts and validates patient factors, generates condition-appropriate medication candidates, ranks options using a hybrid Transformer-GNN model, and returns clinician-reviewable recommendations with grounded evidence, rule-check outcomes, uncertainty indicators, and provenance.



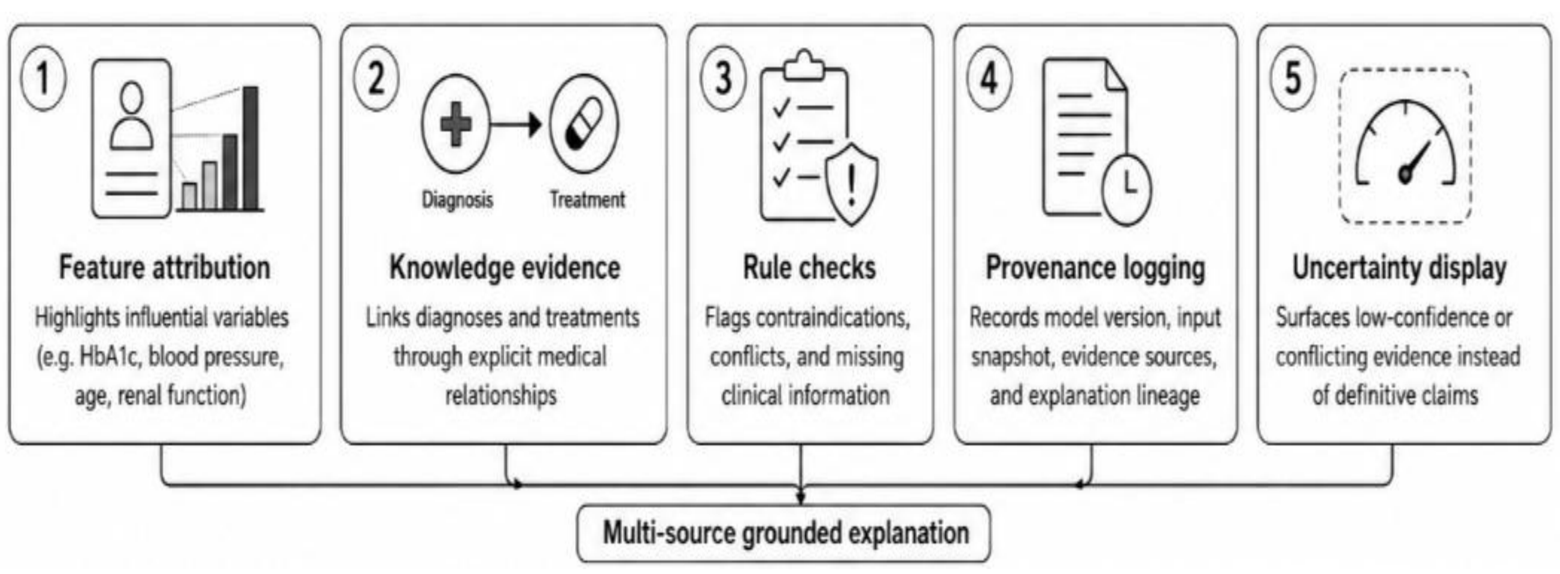
Model Training & Evaluation Pipeline



Integrated Recommendation, Evidence, and Clinician Output

In the resulted research system, clinical dialogue and EHR data are transformed into a structured patient representation, which is then processed directly by a hybrid recommendation module that learns patient-context interactions and medical relationships to rank medication options by relevance; for each ranked option, the system produces grounded evidence by combining model-attribution signals, knowledge-based relational support, and rule-consistency checks, flags missing or conflicting information, and presents a clinician-reviewable output that includes rank, confidence/score, explanation status, uncertainty cues, and provenance so the recommendation remains transparent, auditable, and explicitly supportive of professional judgment rather than a replacement for clinical decision-making.

Explainable AI for Medical Recommendation



Conclusion

This research delivers an explainable conversational medication recommendation framework for clinical decision support, integrating clinical information understanding, structured patient representation, hybrid recommendation modeling, and grounded evidence generation into a single clinician-reviewable workflow. The final system ranks medication options using patient context and medical relationships, and justifies each recommendation through multi-source evidence, rule-consistency checks, uncertainty signaling, and provenance tracking to improve transparency, trust, and practical usefulness in real-world decision-making. It is designed to support healthcare professionals with auditable, evidence-grounded prioritization while preserving clinician authority over all final treatment decisions.